

Contents

An Information-Theoretic Reformulation of Thermodynamic Gravity (Conjecture)	1
Abstract	1
1. Position (B1)	1
2. The gravitational channel and load (B2)	2
3. Newton via Path J/M only (B3)	2
4. Contact with continuum Gravity-from-Entropy (B4)	2
5. The ACD-EW construction as analogy only (B5)	3
6. Falsifiability — the central open question (B6)	3
7. O1 seed — one-slot continuum limit (B7, optional)	3
8. Non-claims (frozen)	4
Changelog	4

An Information-Theoretic Reformulation of Thermodynamic Gravity (Conjecture)

Paper B (conjecture) — **Draft Status:** 2026-07-29 · Preliminary research · position / research-programme paper **Scope contract:** [../DELIVERY_SCOPE.md](#) §2 (may assert B1–B7; non-claims N1–N9) **Depends on:** Paper A [../01-foundations/PAPER_A_computational_entropy.md](#) **Falsifiability verdict:** [../FALSIFIABILITY_L_vs_GR.md](#) **Canonical dynamics:** [../emergent-gravity/master-equation.md](#) · **Newton:** [../emergent-gravity/recoveries/newtonian/README.md](#)

Abstract

Building on the computational-entropy framework of Paper A — where the entropy of a channel’s **output** distribution is a well-defined, Landauer-exact quantity — we ask whether gravity’s known thermodynamic character can be *re-expressed* in this vocabulary. We define a gravitational channel Φ_g and a scalar **computational load** L that reparameterizes proper time, and we present the resulting master equation **honestly as a postulate with a Jacobson-consistent shape**, not as a derivation. Under this postulate the Newtonian weak field is recovered only along an imported route (Clausius → Einstein → Poisson) with one on-shell calibration; we do not derive Newton or Newton’s constant from information alone. We report that our framework and continuum Gravity-from-Entropy reach the same leading Poisson equation (a *weak* agreement forced by both embedding Einstein at low coupling) while diverging at next order. Our central result is therefore **negative-leaning and clarifying**: as constructed the framework is an **information-theoretic reformulation** of thermodynamic gravity, making no sharp prediction distinct from general relativity in any currently accessible regime. We isolate the single term of L — the computational-entropy-production term — where a genuine departure could live, state precisely the missing theorem that would turn it into a prediction, and flag either outcome (reformulation confirmed, or departure found) as acceptable.

1. Position (B1)

This paper is **not** a claim of a new theory of gravity. It is a claim that the *thermodynamic/entropic* content already present in Jacobson’s and Verlinde’s programs can be repackaged in the computational-entropy language of Paper A, and an honest map of exactly how far that repackaging currently reaches. Whether it ever becomes a genuine theory (a falsifiable departure from GR) is **open**; the falsifiability analysis (§6) records the current verdict as *reformulation*.

Every mapping below is tagged **semantic** / **structural** / **constructive**. Type safety is enforced throughout: the load L is a dimensionless scalar; a structure-induced metric G is a tensor; $L \neq G$.

2. The gravitational channel and load (B2)

We model local microstate evolution as a CPTP **gravitational channel** Φ_g ,

$$\rho(\tau + \delta\tau) = \Phi_g[\rho(\tau); g_{\mu\nu}(\rho)], \quad S_c(\Phi_g; \rho) = S(\Phi_g(\rho)),$$

with S_c the von Neumann output entropy (the quantum instance of Paper A’s H_c). The instantaneous processing demand is the **computational load**

$$L(\rho, g) = \beta \frac{E[\rho]}{V\epsilon_0} + \gamma \left| \frac{dS_c}{d\tau} \right|_{\text{reg}} + \delta \frac{S_{\text{boundary}}(\rho)}{S_{\text{BH}}(A)},$$

which clocks proper time via $d\tau = dt/(1 + \alpha L)$, giving the master equation

$$\frac{d\rho}{dt} = \frac{1}{1 + \alpha L(\rho, g)} \mathcal{L}_g[\rho; g_{\mu\nu}(\rho)].$$

Honesty statement (structural). The generator $\mathcal{L}_g$ is *required* to satisfy Clausius $\delta Q = T dS_c$ on local horizons. This is exactly Jacobson’s (1995) condition, from which the Einstein equations follow. The master equation is thus **postulated with the shape that makes it Jacobson-consistent** — it is not independently derived, and on-shell its geometry is GR by construction. What the framework adds beyond GR, if anything, lives entirely in the *clock* L (see §6).

3. Newton via Path J/M only (B3)

We recover the Newtonian weak field along one route, with rigor labels (canonical: [../emerгент-gravity/recoveries/newtonian/README.md](#)):

Step	Content	Rigor
J	Clausius on $\mathcal{L}_g$ Einstein equations	imported theorem (Jacobson) + modeling assumption
—	Einstein weak field $\nabla^2\Phi = 4\pi G\rho_m$	standard GR
M	Load clock $d\tau = dt/(1 + \alpha L)$ matches Newtonian redshift on shell ; $\alpha\beta = 4\pi G/c^4$	calibration

Explicitly not claimed (N5): the withdrawn pointwise identification $\Phi \propto \rho \Rightarrow \nabla^2\Phi \propto \nabla^2\rho$.
Not claimed: a free first-principles value of G , or Newton independent of the Jacobson/Einstein import.

4. Contact with continuum Gravity-from-Entropy (B4)

Comparing our framework to Bianconi’s continuum GfE at low coupling (canonical: [../synthesism6-weak-field-plugtest.md](https://github.com/ianm6/synthesism6-weak-field-plugtest.md)):

Criterion	Outcome
Same leading Poisson $\nabla^2\Phi = 4\pi G\rho_m$	WEAK PASS — both via Einstein/GR
Same derivation mechanism	FAIL
Identifiable $(\alpha_L, \beta_L) = (\alpha_B, \beta_B)$	FAIL / refused
Next-order corrections match	structural FAIL — GfE extras live in the metric EOM $(D_{\mu\nu}, \Lambda_G)$; our γ_L, δ_L live in the load clock unless <i>promoted</i>

The Poisson agreement is **circumstantial co-class with GR**, not evidence that load dynamics equal Bianconi’s relative-entropy equations of motion (non-claim N1).

5. The ACD-EW construction as analogy only (B5)

An earlier warm-up (“Action–Channel Duality, Euclidean”) compares edge-preserving (Perona–Malik) diffusion to isotropic heat on a noisy 1D/2D signal, and finds the edge-preserving reconstructor wins residual on an intermediate time window. We include it **only as an analogy** to Bianconi’s warm-up gradient flow. We note plainly that Perona–Malik edge-preserving diffusion is standard prior art from image processing, that the “field” there is a test signal on a flat lattice (not spacetime), and that the gravity payoff is **analogical, not evidential** (non-claim N7: lattice denoising empirical gravity). This thread is treated as **closed**, not an open crisis.

6. Falsifiability — the central open question (B6)

Since the geometry is GR on-shell, any departure must come from L depending on data not carried by the stress-energy $T_{\mu\nu}$. Term by term (full analysis: [../FALSIFIABILITY_L_vs_GR.md](https://github.com/ianm6/FALSIFIABILITY_L_vs_GR.md)):

1. **Energy term** $\beta E/V\epsilon_0$: a function of $T_{\mu\nu}$, calibrated to GR redshift — **no new prediction**.
2. **Entropy-production term** $\gamma|dS_c/d\tau|$: depends on the *rate of irreversible information processing*, which GR’s proper time does **not** see — **the one candidate departure**.
3. **Boundary term** $\delta S_{\text{boundary}}/S_{\text{BH}}$: only significant near horizons — quantum-gravity regime, not currently testable.

Verdict (recorded): REFORMULATION, with the entropy-production term flagged as the single place a genuine GR departure could live. It is not yet a prediction because (a) degeneracy with an effective stress-energy contribution has not been excluded, and (b) the calibrated magnitude of γ is unknown.

Central conjecture / missing theorem. *Either* exhibit a regime in which $dS_c/d\tau$ is **not** a function of $T_{\mu\nu}$, compute the calibrated γ -residual, and derive a clock-rate prediction distinct from GR ($\rightarrow$ a real experiment: precision timing across strongly irreversible / decoherence-heavy processes at fixed energy density); *or* prove a **degeneracy no-go** showing the three-term load is always reabsorbable into GR + effective source without promotion into the field equations. Either result closes the question and is acceptable.

7. O1 seed — one-slot continuum limit (B7, optional)

The discrete load slots of Paper A ($L_S = H(X | Y)$, extensive; L_E , extensive; L_B , non-additive) are finite classical objects. A first constructive step toward a continuum L is to take **one slot only** — export L_S — on a lattice of local maps with extensive export, and ask whether a scaling limit yields a local export *density*. We present this as an **open first step**, not a completed construction (non-claim N9: IDEM/decay does not yet construct continuum L or G). The target statement: “*there is a scaling limit under which discrete export density converges to a continuum entropy-production density, with stated topology and rate — or a precise obstruction.*”

8. Non-claims (frozen)

N1 master eq continuum GfE · N2 $L \not\equiv G$, $S_c \not\equiv \text{Tr } g \ln G^{-1}$ · N5 no pointwise- ρ Newton · N6 $\gamma_L, \delta_L \neq D_{\mu\nu}, \Lambda_G$ · N7 lattice gravity · N8 GfE not on par with GR · N9 no continuum L/G from IDEM · no free G -from-bits. Full list: [.././synthesis/CURRENT_CLAIMS.md](https://github.com/robertmiller/synthesis/blob/master/CURRENT_CLAIMS.md) §3.

Changelog

Date	Entry
2026-07-29	Initial draft: reformulation stance; channel/load postulate; Path J/M; M6 WEAK PASS/FAIL; ACD-EW as analogy; falsifiability verdict + central conjecture; O1 seed.